

On Rene Descartes's *Discourse on Method*

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Quotations from Descartes's text are taken from the on-line translation by Ian Johnston.]

If one were seeking to select one text from our Liberal Studies Great Books curriculum which first ushers in the modern age, one would have to consider Rene Descartes's *Discourse on Method*, published in 1637, as a uniquely qualified selection. For in this relatively short work Descartes announces an agenda which marks a dramatic and decisive break with past traditions, lays down a project which became (and remains) the central concern of modern Western civilization, and, in the process, sets on the table the most important modern metaphor shaping our attempts to understand nature and ourselves. It's no accident that Rene Descartes has so often been hailed as the first and greatest modern thinker.

One should stress at the outset that, like all great revolutionary writers, Descartes is drawing extensively on the work of his contemporaries, particularly on Galileo (1564-1642), Harvey (1578-1657), and Bacon (1561-1626). But, unlike these well-known natural philosophers, Descartes is both a practising scientist and a first-rate philosopher. He is aware that whatever he has to offer by way of new explanations for natural phenomena must not merely make sense in itself and satisfy experimental testing but also must rest on a coherent method arising from a theory of how the human mind knows, from a theory of knowledge. That is, the new approach to natural science requires a rational justification. This philosophical basis of his scientific writing and the corollaries which arise from it give Descartes's text an enormous importance, over and above any new discoveries or theories he announces.

In this lecture I would like to review some of the more obvious ways in which Descartes' description and justification of his method do indeed mark a revolutionary moment in how we have come to understand what knowledge is and how we acquire and evaluate it. Most of these points are obvious enough from Descartes's text, but, as his heirs who have been raised to believe in his program, we may

not fully grasp just how decisive a turn he is inviting his contemporaries to take.

But first an important caveat. It's important to understand that in the early 17th century there was no clear sense of what the "new" science (or natural philosophy) should or could be. There was not then, as there is now, a general consensus about anything one might call a scientific method or any agreed-upon way to evaluate claims made on the basis of experiments or bold new mathematically-based hypotheses. Thus, while there was certainly a sense of revolutionary change in the air and a growing interest in new ways of looking at natural phenomena, especially with the aid of new machines (telescopes, air pumps, and later microscopes), natural scientists were still in the process of trying to establish and agree about just what their activities involved.

Before reviewing some of the major points Descartes makes, we might first notice the tone of the book, a particularly interesting feature of the argument. Unlike Galileo's frequently satirical voice or Bacon's polemical prose, Descartes strives from the start to reassure us—he is offering, not a blueprint we should all follow, but merely his own personal story, he has no desire for fame nor any radical agenda, especially in any matters which affect his firm religious faith, he's not challenging any important social customs or political arrangements, he's not even sure he wants to write this book, and so on. There is a deliberate sense here of a self-effacing, modest, unambitious, calm, reasonable, and co-operative personality, something very much at odds with the bold and challenging ideas he is setting out in rapid succession.

Of course, this is something of a rhetorical ploy. But we should not therefore dismiss it out of hand as a hypocritical Trojan Horse strategy intended merely to deceive suspicious authorities. For there is no reason to suppose that Descartes is not being perfectly sincere when he affirms his Christian faith or informs us of his intense distaste for personal or political squabbling or proposes a radical scientific hypothesis as a thought experiment rather than the truth of things. Naturally, with the example of Galileo to ponder (as he reminds us), Descartes is well aware of one possible response to what he is proposing, but that does not entitle us to claim (as students sometimes seem fond of doing) that his candour is simply a sham.

In one respect, Descartes's very calm personal tone plays an important part in the argument, of course, since (as we shall see) one of the most distinctive features of what he is proposing is that his method is (he says) something anyone can practise. The book may be his own personal story, but there's an obvious invitation here for others who find what he says agreeable to follow his lead. His decision to write the *Discourse* in French rather than in Latin (the language of the scholars) seems clearly a part of his intention to reassure people about what he is doing and to encourage them to agree with him (and to assist his work).

In some ways, the most devastating aspect of Descartes's argument comes right at the start in the friendly low-key account he provides of his own education. For he makes it clear early on that he is radically dissatisfied with the excellent but conventional education he has received, because it has not answered his desire for truth. He has experienced a thorough immersion in the received wisdom of the ages and the accepted ways of understanding literature, nature, and philosophy, but he wants something none of those disciplines apparently can offer, an assurance of certainty. At the core of his project lies his demand for knowledge which is not subject to disputes or probabilities. Hence, his polite but firm rejection of everything he has been taught.

In making this demand, Descartes is taking issue with the opinions of most of his contemporaries. Many of them continue, some with increasing zealotry, to assert the traditional claims to the truth established by religion and by the long tradition of linking an understanding of nature with divine purposes. Many others, however, including very well educated and reasonable people, have, after more than a century of bloody wars over religious questions, abandoned the notion that such certainty is available in any of the major questions of life and have sought refuge in scepticism or in the developing science of probability. If we live in a world of competing truths and are killing each other over rival interpretations of scripture, we might as well give up a search for Truth (with a capital T) and live quietly, organizing our lives in accordance with the customs of the people around us, the position recommended by Montaigne (1533-1592), or we must seek out and follow what, as best we can ascertain, is most probably true or the course of action which has a better

chance of leading to success (a position most famously publicized by Pascal's Wager, which appeared some years after the *Discourse*: since God either exists and punishes those who disbelieve and rewards those who believe in Him or else He does not exist and there is no afterlife of rewards and punishment, the reasonable person will believe, since he has a lot more to lose from disbelief than from belief).

Descartes's method, then, begins with his rejection of these traditional ways of thinking about the truth. He will adopt a sceptical attitude towards everything he has learned and everything he can perceive with his senses, and he will adopt the customs of the people he lives among, but not with a view to remaining in that state. For he is confident that from this temporary vantage point he can construct something much firmer if he simply looks for it in a new place, namely within himself, within the framework of his own ability to think reasonably and to come to unassailable conclusions about particular things.

In setting up this new method, for all his cautionary words about how it might not suit everyone, Descartes is, in fact, making clear that anyone capable of rational thought will be able, not merely to understand him, but also to follow in his footsteps. His opening comments about the average quality of his own mind are not just false modesty but an important claim for what is to follow. He is thus suggesting two very new features of his method, apart from the fact that it does not require any deference to (or, for that matter, knowledge of) the traditional ways of understanding the world: it is egalitarian, requiring no more reason than most intelligent people possess, and it is self-correcting and progressive, for errors can be dealt with by a better application of the method, and later thinkers can build on the work of earlier ones (false results come from mistakes, not from any inherent limitations in the human mind attributable to original sin). Hence, disputes will be capable of resolution, without the need for interminable arguments about the most basic things, of the sort common to interpretations of Aristotle, for example, or scholarly disputes about ancient texts, in which nothing is ever finally resolved. Such discussions, Descartes points out, may establish a scholar's reputation for learning and make him famous and rich, but they leave us where we always were, in a state of

uncertainty. His method, he is confident, will lead to a progressive improvement in what we know.

Descartes is, of course, best known for his celebrated reflections on what there might be which he can know with certainty and for his first important conclusion: “I think; therefore, I am” (although to describe more accurately what Descartes means, we should probably translate that famous sentence as “I am thinking; therefore, I am,” since the assurance of one’s own existence comes only while the thinking is going on). Hence, while he is capable of doubting or being deceived about nature around him (including his own body), he has something within him, the “I”, which he identifies with the soul, the existence of which is for him absolutely certain, because, even in the process of doubting everything, he cannot deny that he is thinking.

In reviewing Descartes’s description of these first steps in his reflections, we should remember that what he offers in the *Discourse* is merely a summary sketch of the argument he presents in much more detail in the *Meditations on First Philosophy*, published some years later. So those who are tempted to make quick objections should first direct their attention to the later book. What matters here is not so much the philosophical adequacy of the argument but rather the direct effects his reasoning has for an understanding of nature and the method most appropriate for expanding what we know. That said, however, one needs to pay attention to the steps Descartes goes through, because these give a definite shape to the method he is proposing and lead to certain major consequences.

In discussing that famous first conclusion, Descartes establishes his most important criterion for accepting something as true

After that, I considered in general what is necessary for a proposition to be true and certain, for since I had just found one idea which I knew to be true and certain, I thought that I ought also to understand what this certitude consisted of. And having noticed that in the sentence “I think; therefore, I am” there is nothing at all to assure me that I am speaking the truth, other than that I see very clearly that in order to think it is necessary to exist, I judged that I could take as a general rule the point that the things which we conceive very clearly and very distinctly are all true. But that left the single

difficulty of properly noticing which things are the ones we conceive distinctly.

But what, one might ask at once, does he mean by that key phrase “conceive very clearly and very distinctly”? If those are the qualities a thing must possess for it to be true, we need to understand just what they are. About this point there has been much dispute, but in the context of the *Discourse* the words seem to refer to an axiomatic certainty, of the sort we encounter most frequently (as Descartes reminds us) in mathematics, truths which are self-evident and do not admit of reasonable disagreement (like, for example, the claim that things equal to the same thing are equal to each other). And certainly to this point the clarity and distinctness of his thinking seem evident enough, for all the reservations one might have about identifying “I” as a distinct entity coextensive with the soul.

At any rate, this opening move establishes two key elements of Descartes’s theory of knowledge. The first is his radical separation of the thinking ego, the “I,” the soul, from the physical realm of the body and nature. The world, given this metaphor, is fundamentally dualistic, with a spiritual and thinking part within human beings separated from an inert, mechanical world of non-human nature (more about this later). And the second is that a certainty like that provided by mathematical axioms and the deductions of geometry will be the criterion by which we assess what we know.

Descartes then, very quickly and cursorily, faces up to a major corner he seems to have painted himself into. How is he to have any reliable knowledge of the external world, when all his senses are deceptive and the only truth is his own inner process of thinking? How can his knowledge escape total self-consciousness, a thoroughgoing solipsism? The argument (or rather the summary sketch of the later argument) concerning God is the key stage which enables him to take this step.

It is not uncommon for first-time readers of the *Discourse* to find this section rather problematic and to offer the comment that Descartes is here simply placating religious authorities rather than being sincere in his affirmations of belief. Whatever the nature of Descartes’s religious beliefs (and there seems little evidence that his statements about them are not sincere), it’s important to note that the paragraphs about God are an essential part of the philosophical

argument, a necessary logical foundation for the method he is proposing. Without them, the certainty he is seeking would not be available.

The argument for the existence of God is, in part, a traditional one, as Descartes acknowledges. Since he has ideas of perfection and all the flawless qualities of God, he questions where these might have arisen. As a limited and imperfect human being, he does not have those qualities himself. And they cannot have come from sense experience of nature or from outside natural sources (all of which he has discounted). Hence, the idea of such qualities must have come from somewhere else, from some being that manifests these qualities and is of a higher order of goodness than himself, that is, from God. The existence of God is thus necessarily true.¹

The perfection of God then enables Descartes to establish the validity of his principle that what we perceive clearly and distinctly must be true, for “the very principle which I have so often taken as a rule—only to recognize as true all those things which we conceive very clearly and very distinctly—is guaranteed only because of the fact that God is or exists, that He is a perfect being, and that everything which is in us comes from Him. From that it follows that our ideas or notions, being real things which come from God, to the extent that they are clear and distinct, in that respect cannot be anything but true.” Hence, the truth of the clear and distinct ideas we can formulate about the world outside the thinking “I” are underwritten by the existence of a perfect Deity.

To some people this may look suspiciously like circular reasoning: the existence of God is established by the clarity and distinctness of Descartes’s perception of His perfection, and then the perfection of God guarantees the truth of ideas which Descartes perceives clearly

¹Descartes’ proof here has been usefully illustrated with an analogy by John Cottingham. If you found that your ten-year old child had produced a very sophisticated and workable blueprint for, say, a new super-computer, you would naturally assume he or she had received some outside help, since there is, so far as you know, nothing in the child’s education, environment, or character which would possibly have come up with the design. It would be quite reasonable to assume outside assistance or inspiration from some vastly superior intellect. On the other hand, if the child had simply produced a drawing of a box with the words ‘really big and new computer’ written on it, you would not find that problematic. How one assesses Descartes’ argument depends to some extent on which of these two possibilities best fits what he has to say.

and distinctly. Later in the *Discourse*, Descartes quickly dismisses the objection that he uses circular reasoning (using experimental effects to guarantee the truth of his theory of the cause, and then using the cause to guarantee the truth of the effects), but, as I mentioned before, we are being given here merely a summary sketch of the argument, and so if we do have such reservations, we should direct our attention to the case Descartes makes in the *Meditations* (and especially to the *Objections* which Descartes includes and responds to). For our purposes what matters here are some of the more obvious effects of this line of reasoning.

Given what Descartes has said up to this point, certain very revolutionary consequences follow for anyone interested in exploring human knowledge and sorting out what we can and do know from what we are uncertain of. Some of these I have touched on already.

First, according to Descartes's line of reasoning, the world is radically dualistic. The spiritual, knowing world of the "I," the human soul, is set over against the mechanical world of nature (including the human body), which is without any spiritual dimension. The latter operates as a clock. And we understand it, as we do a clock, by using our minds to analyze its parts bit by bit, building our knowledge from simple ideas we are sure about into more complex systems of belief (like the development of geometric theorems).

This metaphor obviously encourages a particular relationship between human beings and nature, giving the knowing mind a pre-eminent position and charging it with the responsibility for finding out about nature. Gone is traditional sense of human beings as privileged participants in nature, with responsibilities for respecting nature either as divine (the classical pagan view) or as a uniquely mysterious creation by God (the traditional Christian view). Descartes's picture of nature provides a license for human beings to probe, explore, experiment—in a word, to tamper with—nature in the search for knowledge, without having to worry about any spiritual qualities in the objects under investigation, because everything outside the human soul is a machine.² This metaphor, more than

²And so Descartes, for example, would consider animals simply complex machines. If a dog howls with pain when it is kicked, that is simply a mechanical response, rather like the gears in a machine squealing when they malfunction. The recent popularity of animal pain as a subject for philosophical speculation is a direct challenge to this old dualism, an attempt to break it down. It is worth noting, too,

anything else, accounts for the astonishingly aggressive attitude Western science quickly developed towards nature.

Later in the *Discourse* the driving motivation for this way of thinking emerges. Descartes wants his method to give us, not merely an understanding of nature, but power over it, something that will make us “as it were, the masters and possessors of nature.” Like Francis Bacon, Descartes wishes to transform the purpose of natural science into something immediately practical, a form of knowledge which people can use effectively to attack and alter nature to suit human purposes (especially in medicine).

Traditionally, of course, the major purpose of natural science, both among the Greeks and the Medieval Christians, was to encourage contemplation. It was, if you like, a form of spiritual discipline or celebration which encouraged in the enquirer a sense of higher moral purpose in the world around us. The very idea of altering nature to fit human desires or seeking to control nature was absurd or impious.

For Descartes and Bacon, this traditional emphasis produced no useful results, nothing which might improve the conditions human beings had to face. A constant preoccupation with the moral purposes of creation produced, in their view, a sterile, unproductive, and disputatious form of knowledge. By setting such concerns to one side and focusing on nature as a soul-less machine whose efficient causes human beings can understand and use, natural science would acquire the knowledge necessary to transform the world for the better.

Descartes is here urging something extremely familiar to us, something we take for granted as a major imperative of our culture. But it was by no means so obvious to his contemporaries, especially

that Descartes is fully aware that there must be some interaction between the soul and the body, since what we think has an effect on how we act. Hence, his later remarks about how the soul is more than a pilot on a ship for “it is necessary that the soul is joined and united more closely with the body, so that it has, in addition, feelings and appetites similar to ours and thus makes up a true human being.” But he has little to offer by way of suggesting how that interaction might take place (elsewhere he locates the place where the soul and the body interact as the pineal gland, because he can find no other use for it). This matter remains a highly contentious matter today, what has come to be called “the problem of consciousness,” especially for those who wish to understand all human experience as merely the result of physical processes.

those who (like many pious scholars antagonistic to Galileo's work) were primarily concerned about what might happen to human thought once the importance of the moral framework of the universe was set aside in the quest for efficient power over nature.

This is not to deny that many modern natural scientists from Descartes's day to the present have been motivated primarily by the wonders of nature, engage in their scientific speculation out of a sense of intense curiosity, and derive contemplative satisfaction from their work. But it is no less true that the major imperative to carry on science so intensely in schools and the research establishments has increasingly become the desire to gain more power over nature, even at the expense of that wonder. Many writers have acknowledged that the modern project launched by Descartes and others has, in fact, gradually emptied the world of wonder, for we have so many reasonable explanations for natural phenomena and have placed so much of nature directly under our own control, that we are seldom personally confronted with nature as a mysterious presence.

What Descartes is proposing here also has important consequences for the vital role mathematics must play in our understanding of nature. For the clarity and directness which, thanks to God, reveal the truth to us are, above all, conveyed in mathematical deductions. Hence, this *Discourse* launches the strong demand that modern science must follow mathematical logic, and the truth of its claims emerges from the mathematical foundations upon which those are based. Science, in other words, needs to rely upon equations rather than verses cited from scripture or traditional interpretations of Aristotle.

This emphasis is significantly different from early 17th century science in England, where, under the energetic leadership of Francis Bacon, the stress is much more on experimental evidence, the collection of observed facts, and the inductions one might draw from repeated observation and testing. And for some time, there was a lively dispute between English science (based on experiments) and Continental science (based on mathematics). In fact, however, one should not over-emphasize these differences. Descartes makes clear in the *Discourse* how important experiments are, and Bacon repeatedly called attention to the imperfections of sense experience (all the more

evident when frequently defective machinery was involved, like imperfect lenses in telescopes and leaky air pumps).

The major synthesis between the two approaches to natural science occurred in the work of Isaac Newton (1643-1727), whose work combined mathematically-based hypotheses with key experiments to test the explanatory power of the equations.

Once he has established the basis of his method, Descartes then quickly offers a summary list of what he turned his attention to, the natural phenomena which he feels his method has enabled him to understand. In the midst of this section, he puts on the table an idea which was later to have the most revolutionary implications for our understanding of nature and ourselves, that is, a historical approach to understanding nature scientifically. It's worth having a close look at this suggestion, particularly for anyone who is going on to follow what later thinkers will do with what at this stage is little more than a tentative hypothesis.

What if, Descartes suggests "God now created somewhere in imaginary space enough material to compose [a new world], and if He set in motion, in a varied and disorderly way, the various parts of this material, so that it created a chaos as confused as poets could make it, and then afterwards He did nothing other than lend His ordinary help to nature and allow it to act according to the laws which He established." What if, in other words, we were to seek to understand the world as something with a unique history, a series of different stages through which it passed, under the guidance of divinely ordained natural laws, from disorganized materials into the ordered structure we see all around us?

This is, for Descartes, an extremely bold and potentially dangerous idea. For at the centre of a Christian understanding of nature had long been the insistence, based on scripture, that God created the world and everything in it in its present state. Hence, the earth does not have a history, not in the sense of a unique linear development out of something very different from what it is now. Descartes's proposal suggests that the only permanent things are the laws God established to allow nature to act as He wanted it to through various stages and that we can come to a useful understanding of what is in the world by seeing it as a something with a particular history.

The second doctrinal problem this suggestion has always run into is that it imposes restrictions on God. If matter always obeys the laws God has established to guide its development, then He is not free to intervene in unexpected and miraculous ways. God is, in effect, the servant of His own laws, not totally free. This point has always been the major objection of Christian thinkers to the often popular view called Deism (which corresponds roughly with what Descartes is proposing: God created the original matter, established the natural laws governing its development, and does not intervene).

Newton's opposition to this view of the world is well known. For him, God established the universe in its present form, and the only possible changes came from unexpected "miraculous" divine interventions. Nonetheless, within a century of Newton's great work, natural scientists (including Immanuel Kant) were writing accounts of the historical development of the earth and of the universe itself.

The dangers in this approach stem not merely from its challenge to orthodox belief. Any historically based understanding of the world carries a latent political message, as well, namely that the existing state of things is not something permanently ordained but rather the result of a historical process (which, presumably, is still going on). Hence, a historical approach brings with it potentially explosive political implications (something which is exploited dramatically by Jean Jacques Rousseau in the 18th century). Eventually, of course, this way of understanding the past leads directly to two of the most revolutionary thinkers of all, Charles Darwin and Friedrich Nietzsche, who, in effect, undercut all claims to the permanence of anything.

Descartes presents this historical proposal only tentatively, as a thought experiment, something useful for our thinking about the world, but not the truth of what really occurred.

However, I don't want to suggest from all these things that this world was created in the fashion which I was proposing. For it is much more probable that God made the world from the beginning just what it had to be. But it is certain, and this is an opinion commonly received among theologians, that the actions by which God now preserves the world are exactly the same as the method by which He created it, in such a way

that even if He did not give it at the start any form other than a chaos, providing that He had first established the laws of nature and had given His assistance, so that it would act as it usually does, we can believe, without denying the miracle of creation, that because of these facts all purely material things would have been able, over time, to become the way we now observe them, and their nature is much easier to conceive when one sees them born gradually in this way than if one thinks of them only as made all at once.

This stance is not necessarily, as with Rousseau (in his *Second Discourse*), an obvious rhetorical ploy to fend off religious authorities, but a standard and acceptable way to offer a scientific speculation without contradicting orthodox religious doctrine. The introduction to Copernicus's revolutionary book on the sun-centred solar system (a letter written anonymously by the Lutheran theologian Andreas Osiander and put in the book to replace Copernicus's original Preface) made exactly the same claim (in 1543) and was, many assert, the reason the book was not condemned. Nonetheless, the mere existence of such "thought experiments" is enough to get people thinking about them as possible versions of what really exists.

Descartes concludes his *Discourse* by inviting his readers to assist him in his endeavours (this plea may well be the main reason for his publishing it in the first place). What's striking about this section is the way in which it indicates how Descartes sees his new method as a cooperative enterprise, something to which the general public could and should contribute, both with money and with ideas (objections, new experiments, and so on). Clearly, he wants his new method to generate discussion and significant interaction. Natural science, he is asserting, is not something to be restricted to the schools but carried on as a widely publicized activity, because its progressive character requires and makes possible contributions from all quarters.

This, too, marks something significantly new—an opening up of natural science to everyone who wishes to contribute. It's not surprising that the years immediately following the publication of the *Discourse* (and other works urging similar public communication in different countries of Europe) saw the establishment of officially sanctioned scientific societies, whose major purpose was to fund and carry out experiments and publicize the results. This quickly became

an international endeavour, often fuelled by intense rivalries which had a catalytic effect on the development of modern science, until we reached the condition we are all too familiar with, modern science and its technological offshoots as the dominating activities of modern society, the lifeblood of our huge public universities and any number of research institutes.